

Deep Learning: Theory and Practice

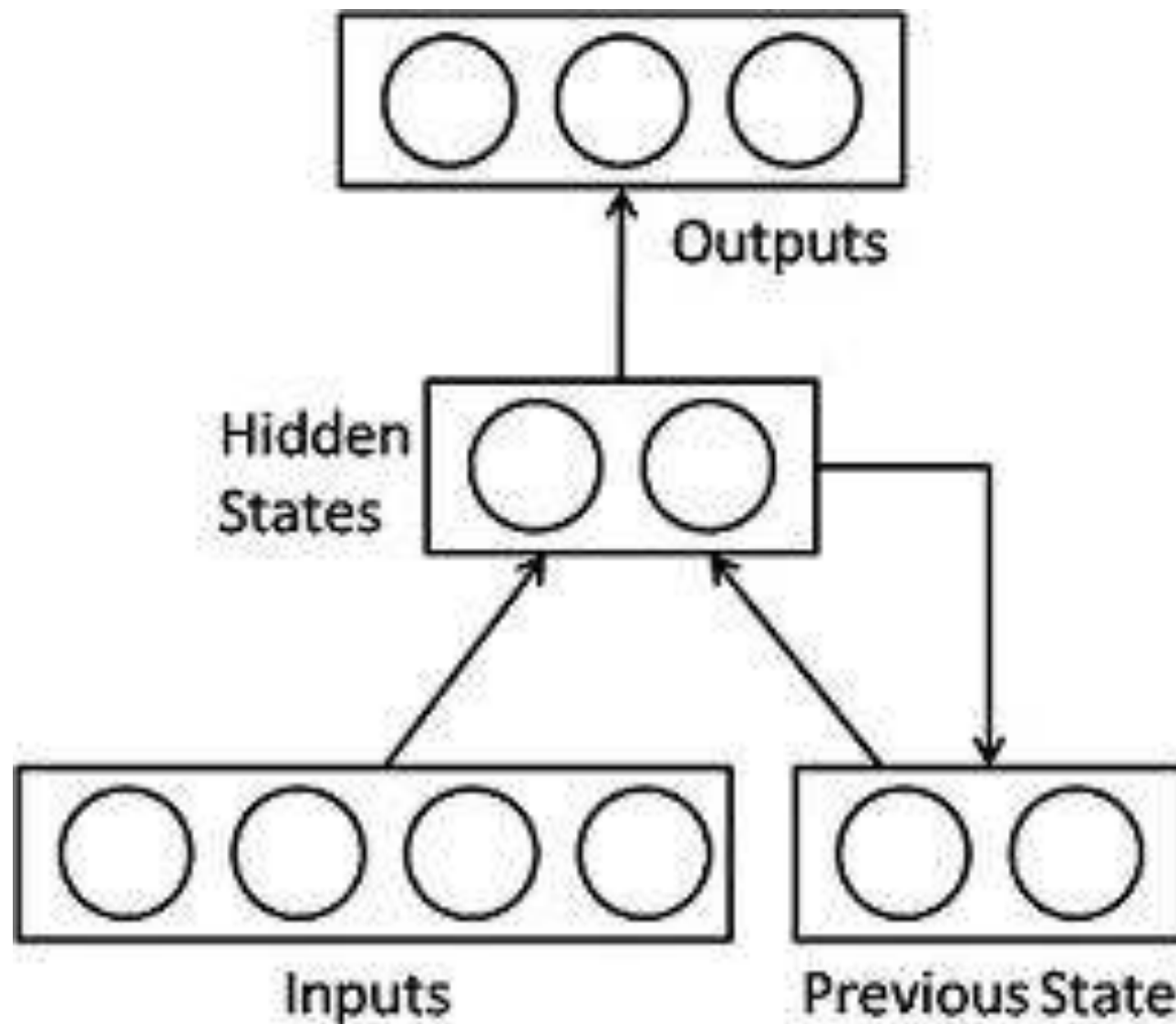
Recurrent Neural Networks

04-04-2018

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Recurrent Operations



Feedback

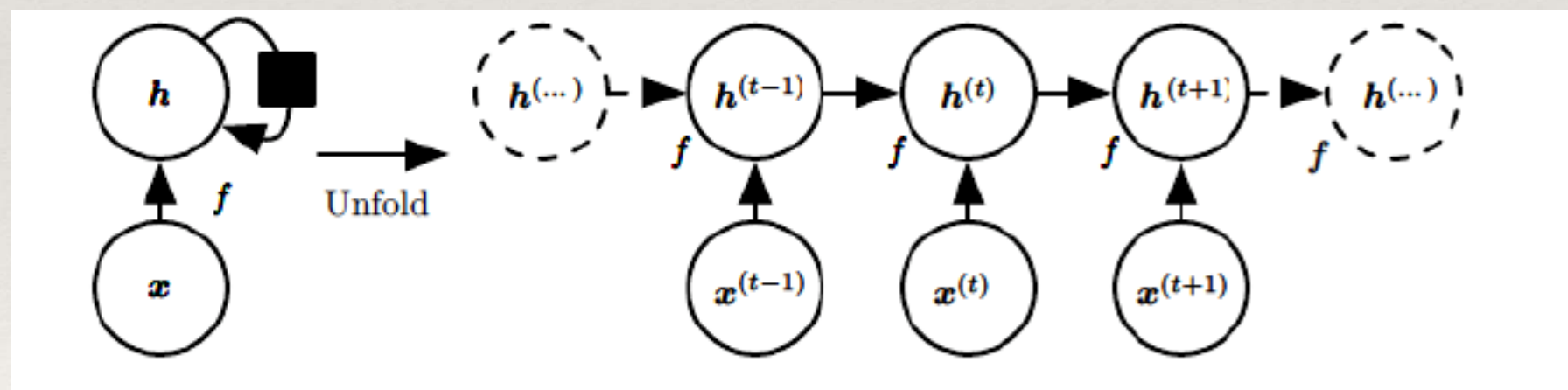
Recurrent Networks

$$\mathbf{s}^{(t)} = f(\mathbf{s}^{(t-1)}; \boldsymbol{\theta}),$$

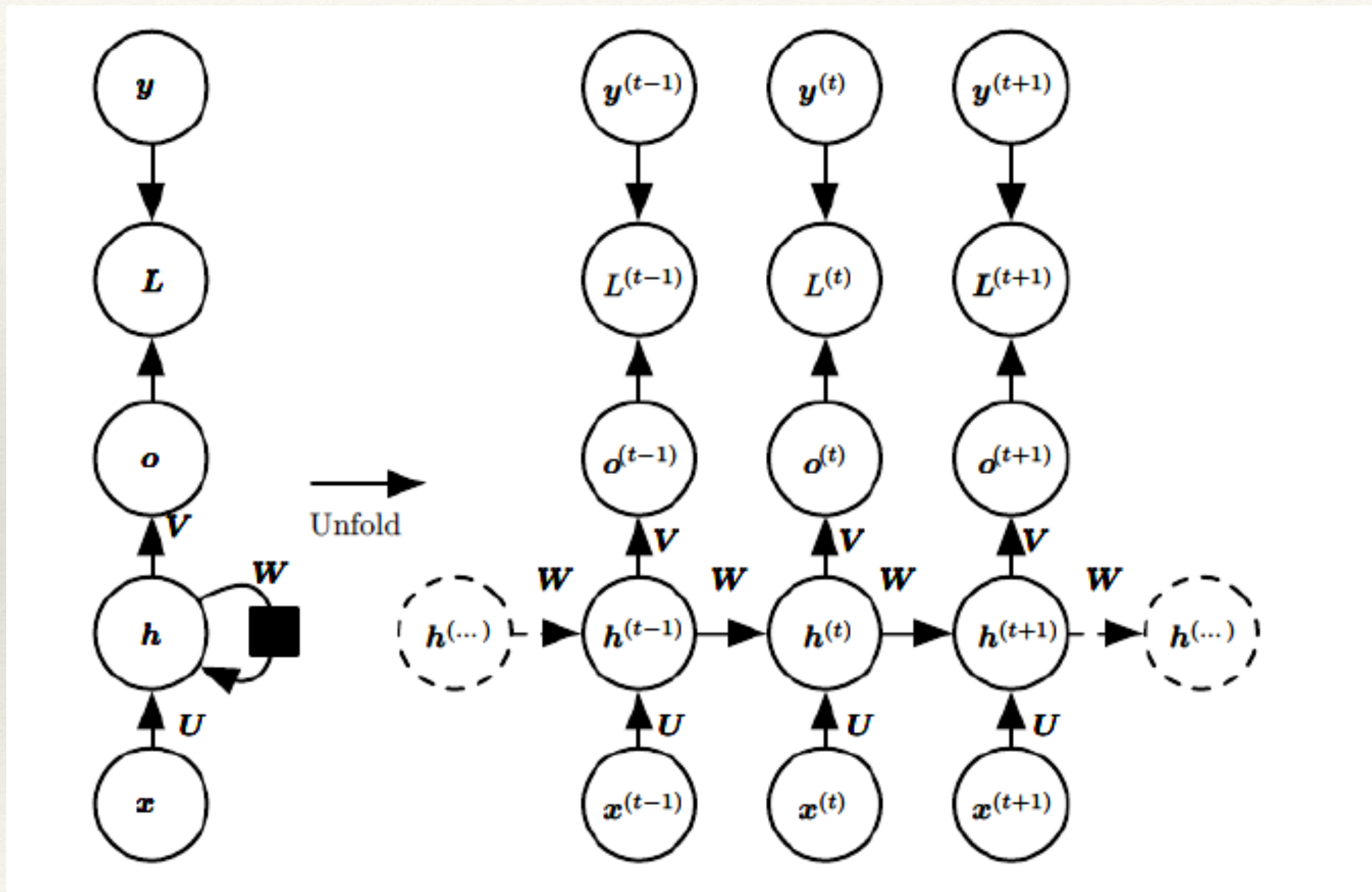
$$\begin{aligned}\mathbf{s}^{(3)} &= f(\mathbf{s}^{(2)}; \boldsymbol{\theta}) \\ &= f(f(\mathbf{s}^{(1)}; \boldsymbol{\theta}); \boldsymbol{\theta})\end{aligned}$$

$$\mathbf{s}^{(t)} = f(\mathbf{s}^{(t-1)}, \mathbf{x}^{(t)}; \boldsymbol{\theta}),$$

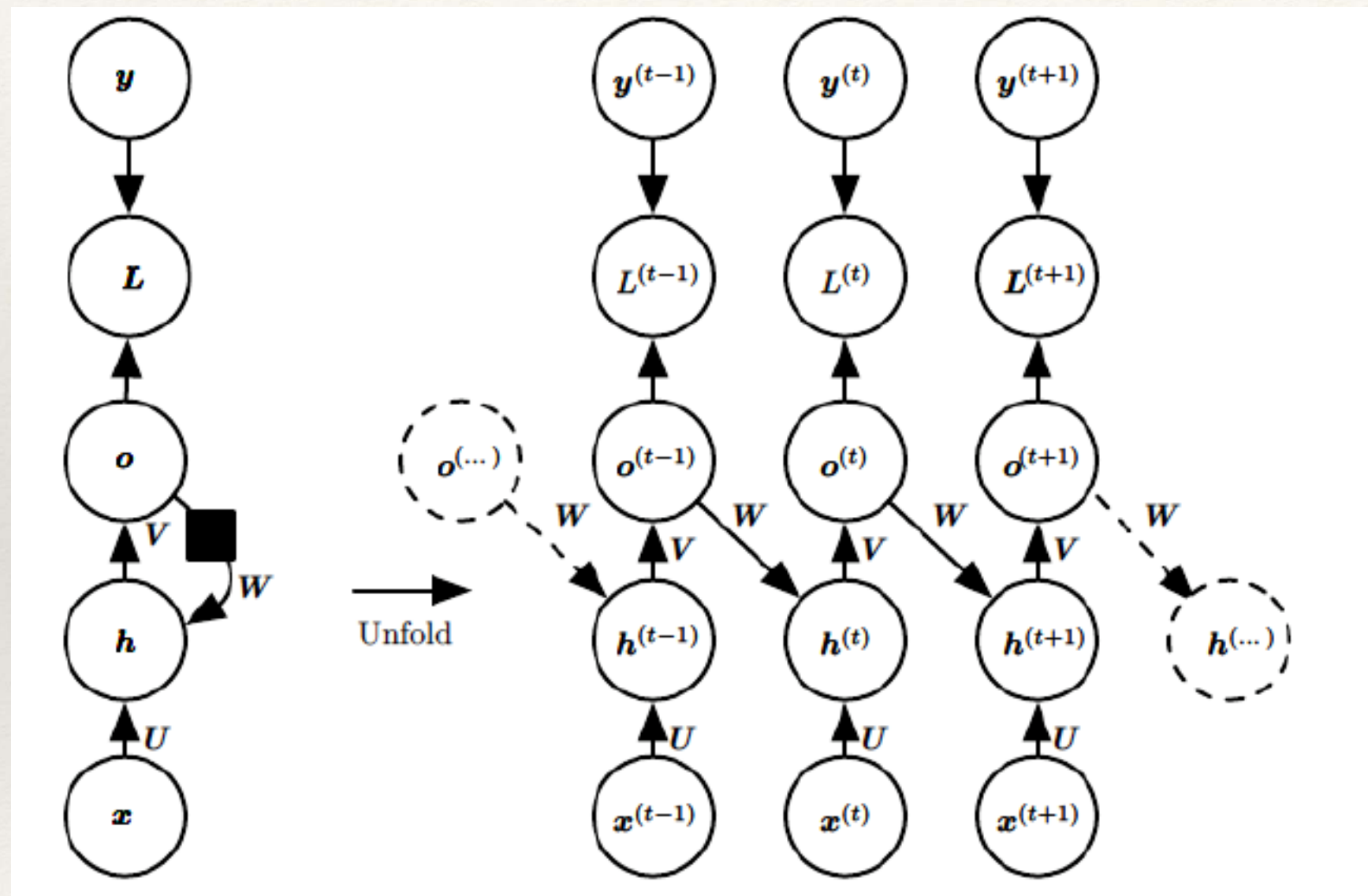
$$\mathbf{h}^{(t)} = f(\mathbf{h}^{(t-1)}, \mathbf{x}^{(t)}; \boldsymbol{\theta}),$$



Recurrent Networks

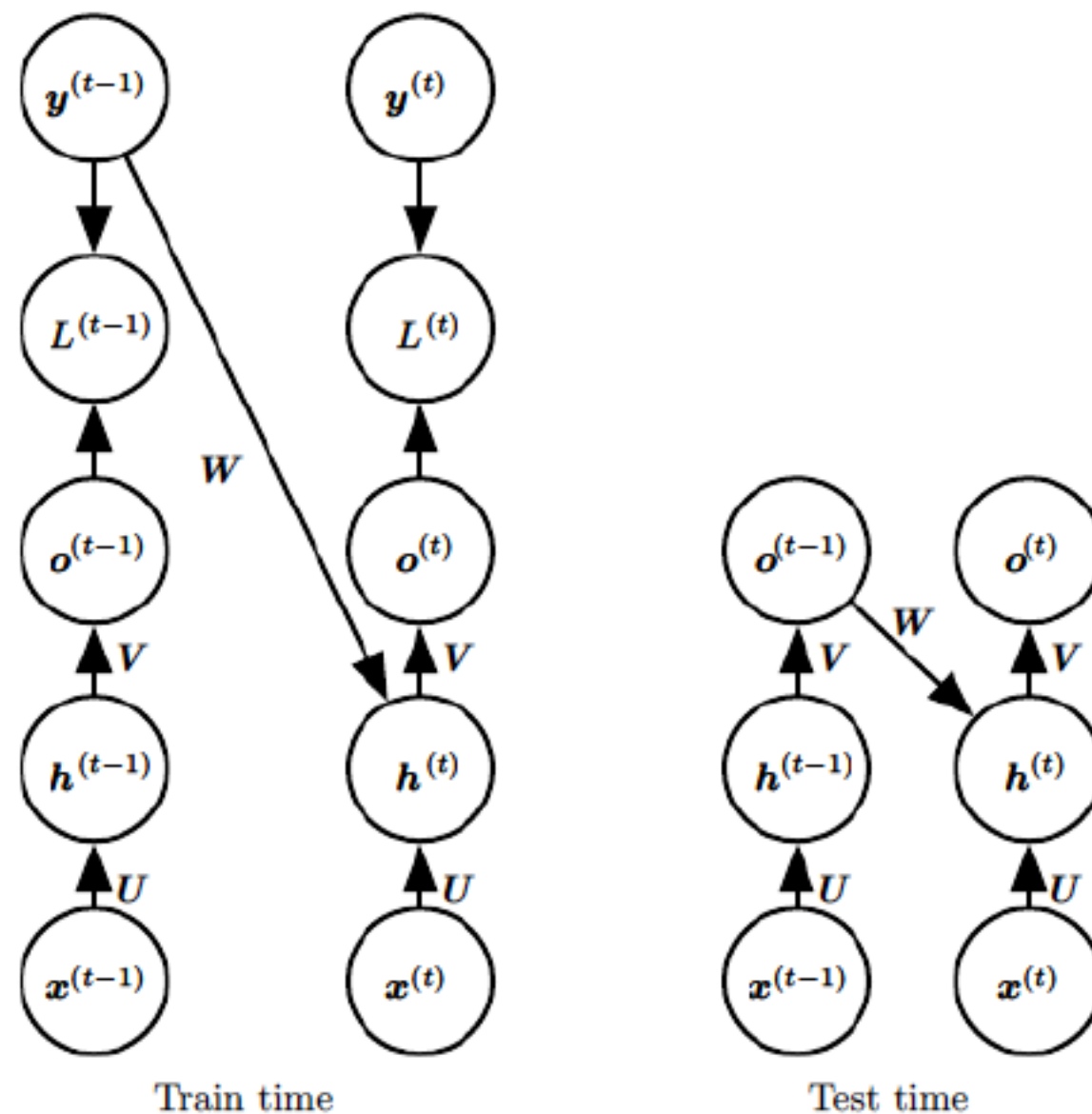


Recurrent Networks



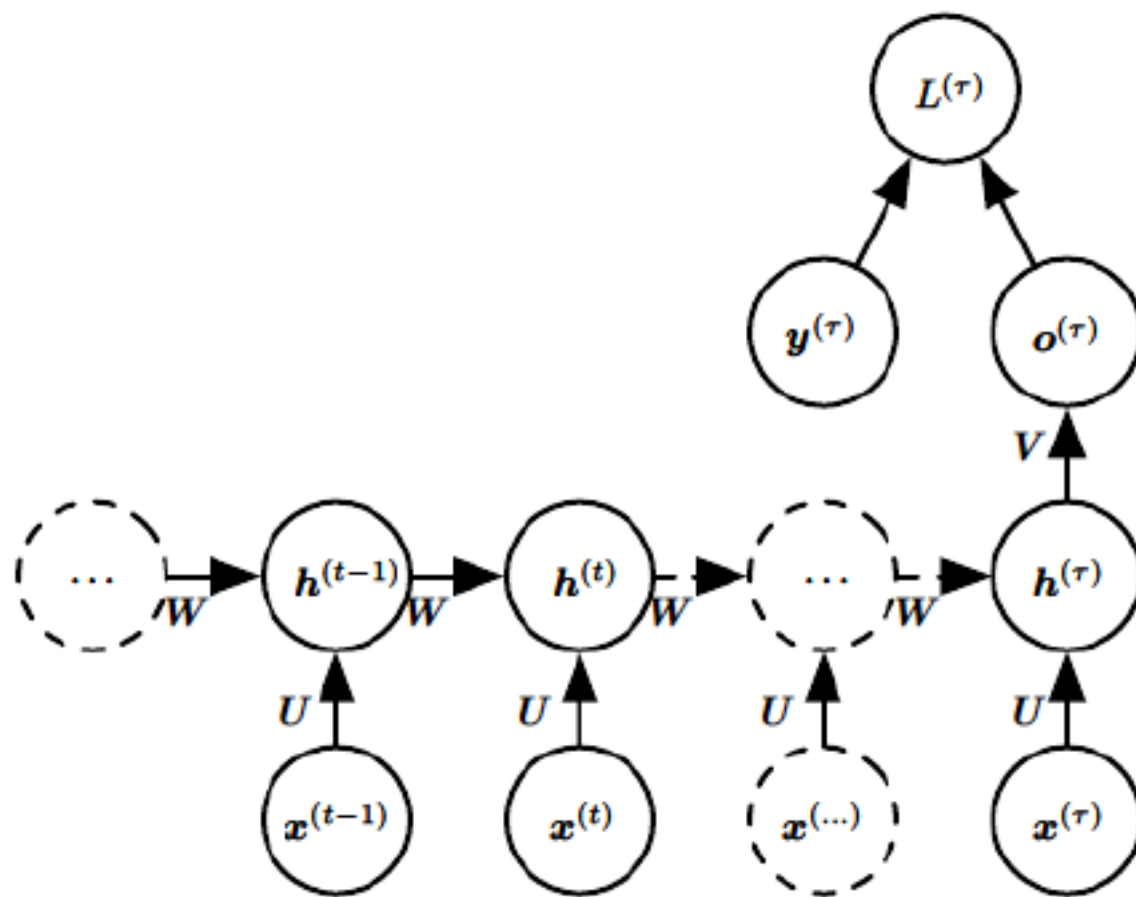
**Teacher
Forcing Networks**

Recurrent Networks



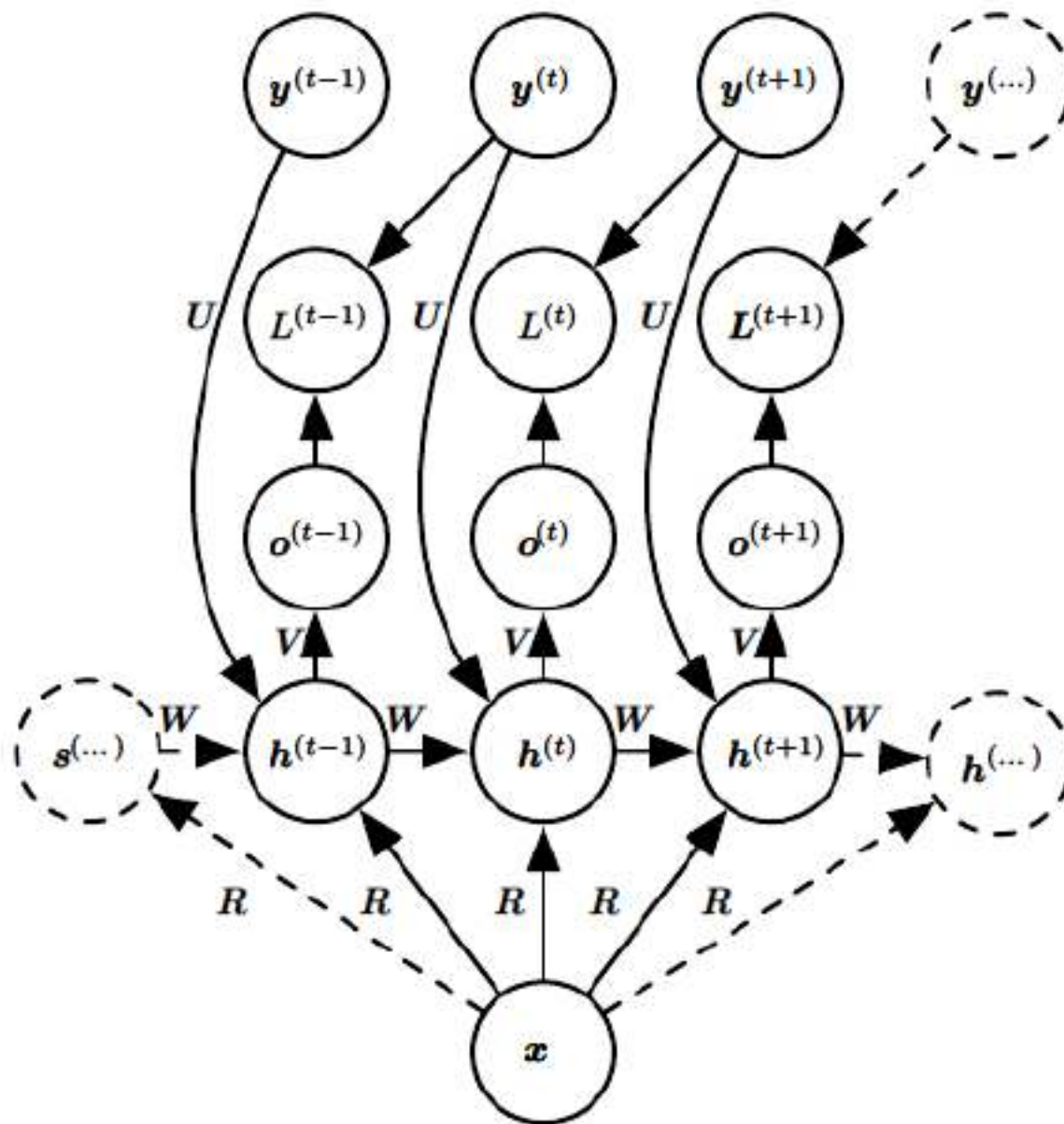
**Teacher
Forcing Networks**

Recurrent Networks



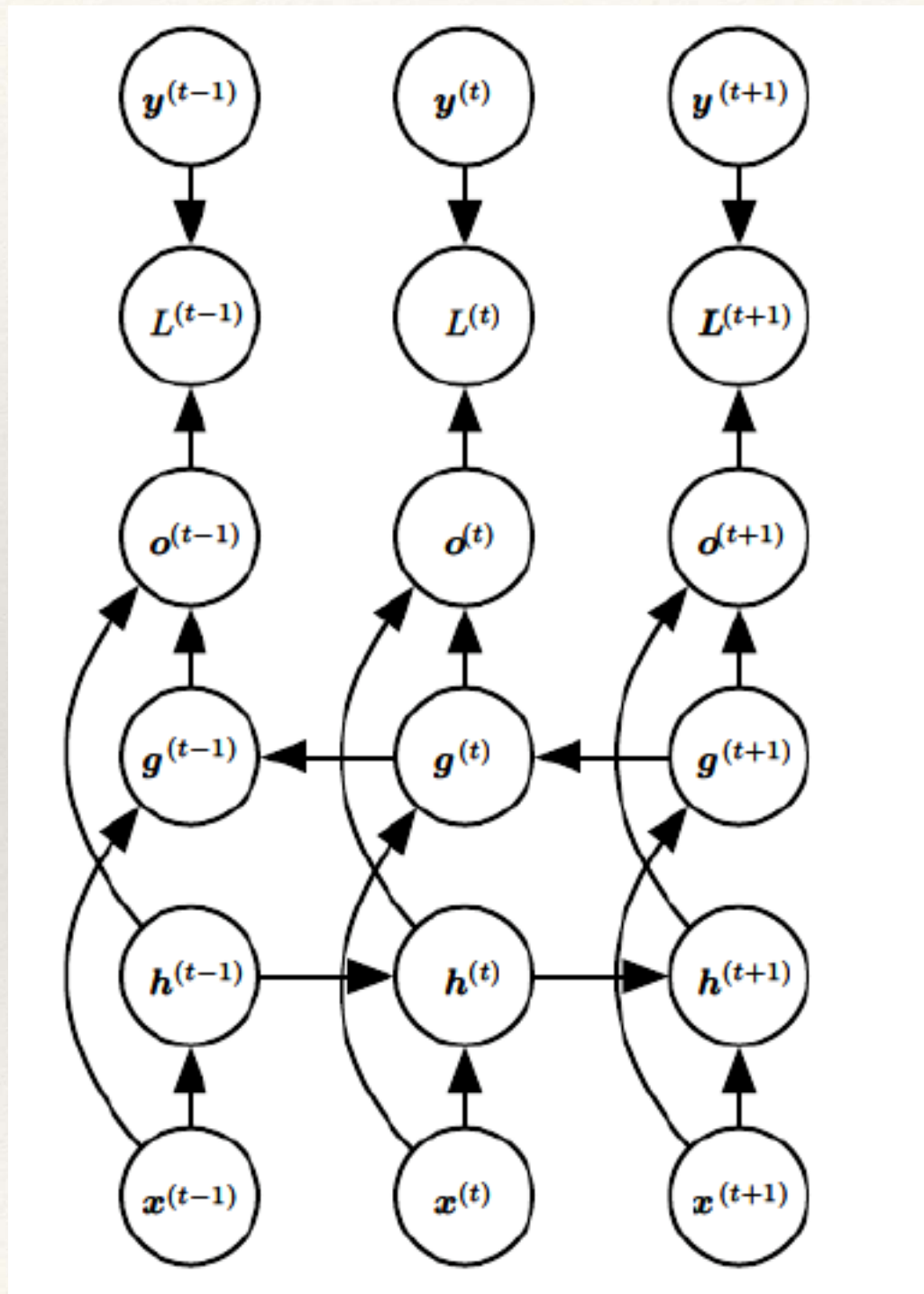
**Multiple Input
Single Output**

Recurrent Networks



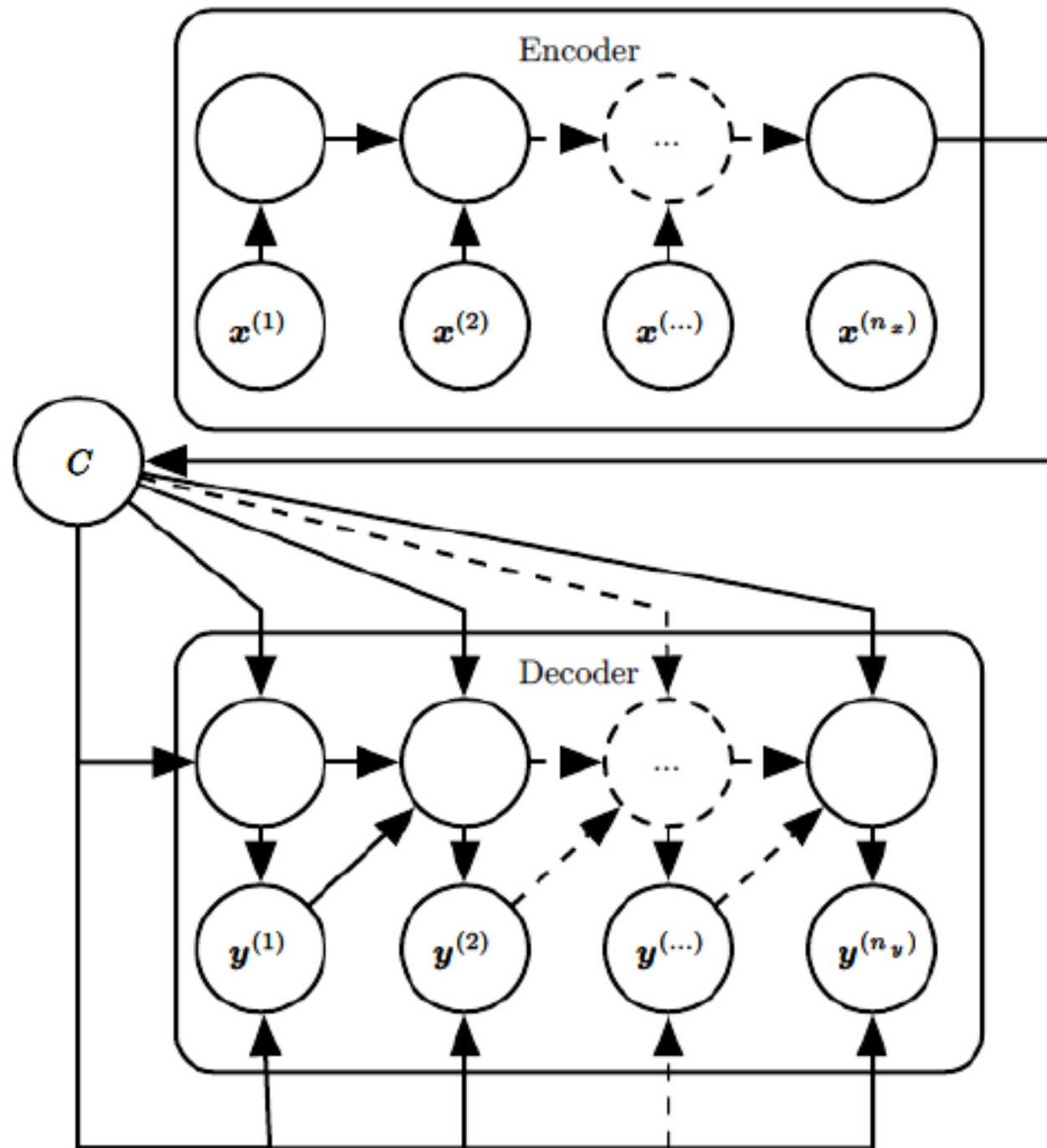
**Single Input
Multiple Output**

Recurrent Networks



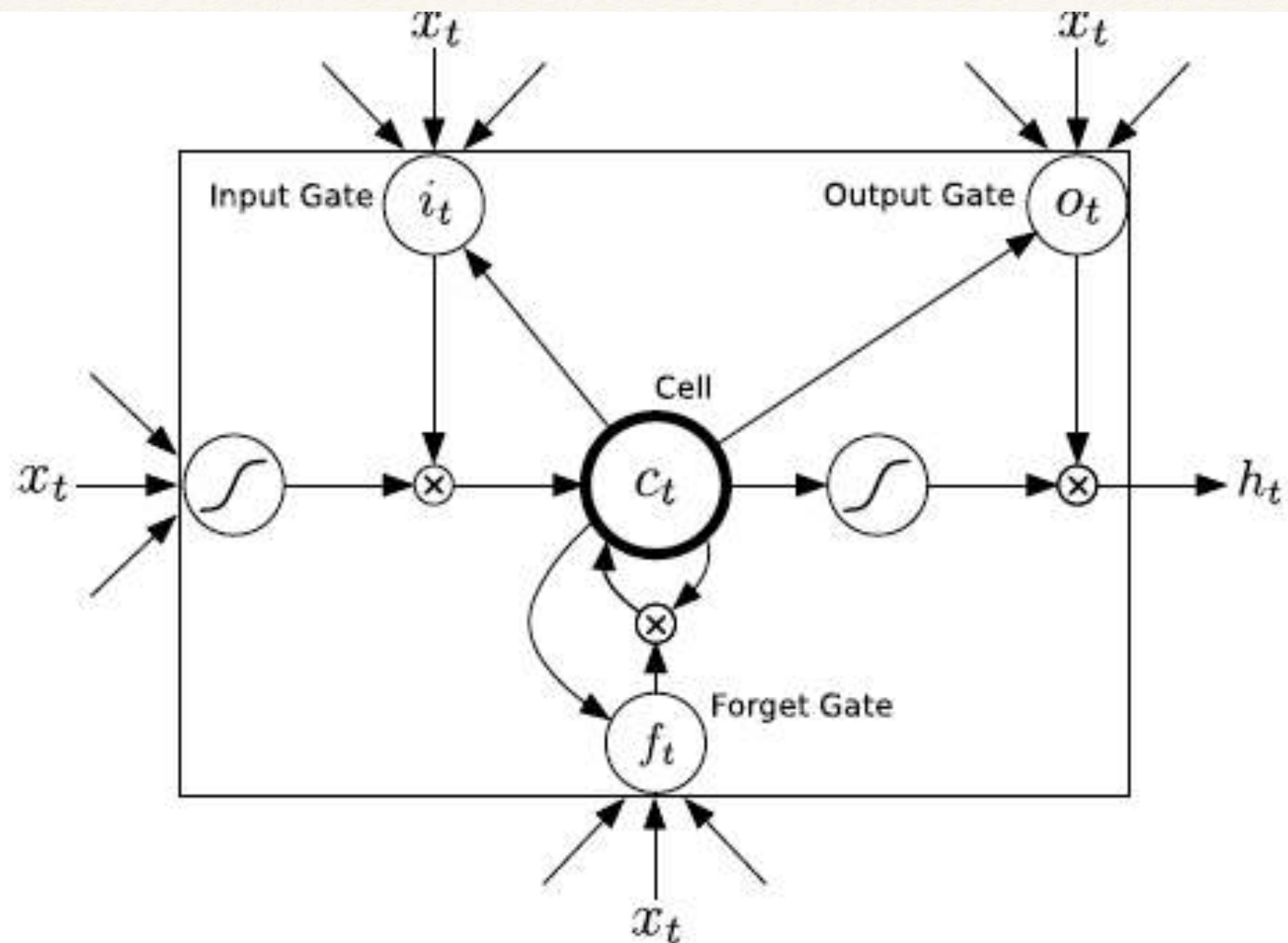
**Bi-directional
Networks**

Recurrent Networks



**Sequence to
Sequence
Mapping Networks**

LSTM



LSTM

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f)$$

$$c_t = f_t c_{t-1} + i_t \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_t + b_o)$$

$$h_t = o_t \tanh(c_t)$$

Bidirectional LSTMs

